

# Package ‘ndm’

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**Title** Neural Disease Modeling Software

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**Description** Provides an R-first interface for neural disease modeling with a reticulate-managed JAX backend. The package includes backend bootstrap helpers, built-in epidemiological model specifications, TFRecord-consuming data interfaces, package-managed runtime helpers, and package-native orchestration helpers for model execution.

**Depends** R (>= 4.1.0)

**Imports** data.table,  
ndmdatasets,  
rlang,  
reticulate,  
utils

**Suggests** fastmatch,  
progress,  
rapply,  
testthat (>= 3.0.0),  
yaml,  
zip,  
zoo

**License** AGPL-3

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**Config/testthat/edition** 3

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|             |                                         |
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| ndm-package | <i>Neural Disease Modeling Software</i> |
|-------------|-----------------------------------------|

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## Description

ndm is an R-first package for neural disease modeling with a reticulate-managed JAX backend. The package currently provides backend setup, epidemic model spec import/export, TFRecord readers, package-managed runtime helpers, package-native runtime/model helpers, package-native run APIs, and artifact helpers for saving and restoring trained models.

## Details

The current implementation preserves both historical model families, "DecoderOnly" and "NeuralODE", while defaulting to "DecoderOnly". The supported backbone is the transformer path only. Latent attention and Mamba are intentionally excluded.

The package treats the structured `ndm_model_spec` object as the canonical internal representation and supports `.tex` import/export for compatibility with the original epidemic specification workflow.

For executable workflows rather than dry-run previews, install the helper packages listed under Suggests, initialize the backend with `ndm_initialize_backend`, and align that backend with `NDM_SOFTWARE_CONDA_ENV` or `NDM_CONDA_ENV` when using the package-native runner helpers. When both variables are unset, the runner layer currently falls back to `jax_cpu` on macOS and `ndm_software_env` elsewhere.

## Key functions

- `ndm_create_config`
- `ndm_build_backend`
- `ndm_initialize_backend`
- `ndm_model_spec`
- `ndm_model_spec_from_tex`
- `ndm_model_spec_to_tex`
- `ndm_read_tfrecord_dataset`
- `ndm_load_tfrecord_bundle`
- `ndm_prepare_runtime`
- `ndm_prepare_data`
- `ndm_build_model`

- ndm\_train
- ndm\_predict
- ndm\_loss
- ndm\_fit
- ndm\_create\_real\_run\_config
- ndm\_run\_real
- ndm\_run\_sim
- ndm\_run\_multidisease
- ndm\_save\_model
- ndm\_load\_model
- ndm\_resume\_training

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ndm\_bootstrap\_sim\_tfreords

*Bootstrap canonical simulation TFRecords by BaseID*

---

**Description**

This helper materializes canonical simulation TFRecords without going through the legacy ‘outer’ scheduling path. It reads the simulation grid in raw CSV or data-frame order, builds one canonical write plan row per ‘BaseID’, and optionally writes those TFRecords with a per-‘BaseID’ cross-process lock.

**Usage**

```
ndm_bootstrap_sim_tfreords(
  project_root = getwd(),
  analysis_name = "BigSimsLatest",
  grid = NULL,
  grid_file = file.path("Data", "RunGrids", "SimGrids", sprintf("SimGrid_%s.csv",
    analysis_name)),
  base_ids = NULL,
  tfrecord_dir = file.path("Data", "RunTFRecords", "SimTFRecords", analysis_name),
  overwrite = FALSE,
  dry_run = FALSE
)
```

**Arguments**

|               |                                                                                |
|---------------|--------------------------------------------------------------------------------|
| project_root  | Working project directory used to resolve relative grid and TFRecord paths.    |
| analysis_name | Analysis label used to derive default grid and TFRecord paths.                 |
| grid          | Optional in-memory simulation grid. When supplied, ‘grid_file’ must be ‘NULL’. |
| grid_file     | Optional CSV path for the simulation grid.                                     |

|              |                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| base_ids     | Optional integer vector of canonical 'BaseID' values to materialize. When 'NULL', bootstrap all 'BaseID's in raw grid order of first appearance. |
| tfrecord_dir | Output directory for canonical TFRecords.                                                                                                        |
| overwrite    | Logical scalar controlling whether existing canonical TFRecords should be re-generated.                                                          |
| dry_run      | Logical scalar indicating whether to return the canonical write plan without writing TFRecords.                                                  |

### Value

A data frame with one row per canonical 'BaseID' and columns 'BaseID', 'selected\_rows', 'canonical\_row', 'artifact\_n\_samples\_train', 'train\_file', 'inference\_file', and 'status'.

---

|                 |                                                                   |
|-----------------|-------------------------------------------------------------------|
| ndm_build_model | <i>Build and train models with package-managed runtime stages</i> |
|-----------------|-------------------------------------------------------------------|

---

### Description

These wrappers layer a small R API over package-managed Phase 1 model build and training stages.

### Usage

```
ndm_build_model(
  runtime_env,
  model_type = c("DecoderOnly", "NeuralODE"),
  model_spec = NULL,
  backbone = "transformer",
  runtime_globals = list()
)

## S3 method for class 'ndm_model'
print(x, ...)

ndm_train(x, run_define = TRUE, run_loop = TRUE)

## S3 method for class 'ndm_trained_model'
print(x, ...)

ndm_fit(
  config = ndm_create_config(),
  model_spec = NULL,
  data_generator = c("sim", "real", "multidisease"),
  runtime_globals = list(),
  data_globals = list(),
  build_globals = list(),
  train_globals = list(),
  run_define = TRUE,
  run_loop = TRUE
)
```

**Arguments**

|                              |                                                                                                                                                                                                                          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>runtime_env</code>     | Runtime environment containing loaded runtime helpers and data globals.                                                                                                                                                  |
| <code>model_type</code>      | Model family to build. Either <code>"DecoderOnly"</code> or <code>"NeuralODE"</code> .                                                                                                                                   |
| <code>model_spec</code>      | Optional <code>'ndm_model_spec'</code> object used to override the model TeX specification supplied to the runtime model builder.                                                                                        |
| <code>backbone</code>        | Backbone family. Phase 1 supports <code>"transformer"</code> only.                                                                                                                                                       |
| <code>runtime_globals</code> | Named list of additional globals assigned during the relevant runtime stage. <code>'ndm_build_model()'</code> uses them before building the model, while <code>'ndm_fit()'</code> uses them before sourcing the runtime. |
| <code>x</code>               | Either an <code>'ndm_model'</code> , an <code>'ndm_trained_model'</code> , or a prepared runtime environment, depending on the function being called.                                                                    |
| <code>...</code>             | Unused; included for S3 method compatibility.                                                                                                                                                                            |
| <code>run_define</code>      | Logical scalar indicating whether the training definition script should be sourced.                                                                                                                                      |
| <code>run_loop</code>        | Logical scalar indicating whether the training loop script should be sourced.                                                                                                                                            |
| <code>config</code>          | An object of class <code>'ndm_config'</code> , usually created by <code>'ndm_create_config()'</code> .                                                                                                                   |
| <code>data_generator</code>  | Which local data generator to source before calling <code>'ndm_build_model()'</code> inside <code>'ndm_fit()'</code> .                                                                                                   |
| <code>data_globals</code>    | Named list of globals assigned before sourcing the data generator in <code>'ndm_fit()'</code> .                                                                                                                          |
| <code>build_globals</code>   | Named list of globals assigned before building the model in <code>'ndm_fit()'</code> .                                                                                                                                   |
| <code>train_globals</code>   | Named list of globals assigned immediately before training in <code>'ndm_fit()'</code> .                                                                                                                                 |

**Details**

These functions assume the backend has already been initialized with `'ndm_initialize_backend()'` and that `'runtime_env'` already contains the upstream runtime and data globals required by the selected workflow. `'ndm_train()'` requires `'rrapply'`; it also requires `'zip'` when checkpointing is enabled and `'zoo'` for multidisease training. Simulation paths inherit the `'progress'` and `'zoo'` requirements documented on `[ndm_prepare_runtime()]`.

**Value**

`'ndm_build_model()'` returns an object of class `'ndm_model'`. `'ndm_train()'` and `'ndm_fit()'` return an object of class `'ndm_trained_model'`.

`'print.ndm_model()'` prints a brief summary of an `'ndm_model'` object and invisibly returns `'x'`.

`'print.ndm_trained_model()'` prints a brief summary of an `'ndm_trained_model'` object and invisibly returns `'x'`.

**Examples**

```
cfg <- ndm_create_config()
spec <- ndm_model_spec()
## Not run:
runtime_env <- ndm_prepare_runtime(cfg)
ndm_prepare_data(runtime_env, generator = "sim")
model <- ndm_build_model(runtime_env, model_spec = spec)
trained <- ndm_train(model)

## End(Not run)
```

---

|                   |                                                    |
|-------------------|----------------------------------------------------|
| ndm_check_backend | <i>Provision and initialize the Python backend</i> |
|-------------------|----------------------------------------------------|

---

## Description

These helpers manage the reticulate-backed Python environment used by ndm. Use `ndm_build_backend()` to provision dependencies, `ndm_check_backend()` to verify availability, and `ndm_initialize_backend()` to import the active JAX stack into R.

## Usage

```
ndm_check_backend(
  conda_env = "ndm_software_env",
  conda = "auto",
  modules = c("jax", "numpy", "optax", "equinox", "diffrax")
)

ndm_build_backend(
  conda_env = "ndm_software_env",
  conda = "auto",
  python_version = "3.13",
  include_tensorflow = TRUE,
  extra_packages = c("optax", "equinox", "diffrax")
)

ndm_initialize_backend(
  conda_env = "ndm_software_env",
  conda = "auto",
  conda_env_required = TRUE,
  float_type = c("32", "64"),
  import_tensorflow = TRUE
)

ndm_backend_modules()
```

## Arguments

|                                 |                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------------------|
| <code>conda_env</code>          | Name of the conda environment that should contain the JAX runtime.                                        |
| <code>conda</code>              | Conda executable to use. <code>"auto"</code> delegates discovery to reticulate.                           |
| <code>modules</code>            | Character vector of Python module names that must be importable for the backend to be considered healthy. |
| <code>python_version</code>     | Python version requested when creating the conda environment.                                             |
| <code>include_tensorflow</code> | Logical scalar indicating whether TensorFlow should be installed alongside JAX.                           |
| <code>extra_packages</code>     | Additional Python packages to install with <code>'pip'</code> .                                           |
| <code>conda_env_required</code> | Passed through to <code>'reticulate::use_condaenv()'</code> as the <code>'required'</code> flag.          |
| <code>float_type</code>         | Floating point precision used when configuring JAX. Either <code>"32"</code> or <code>"64"</code> .       |

```
import_tensorflow
```

Logical scalar indicating whether TensorFlow should be imported when it is available in the selected environment.

## Details

When you want backend helpers and the package-native `'ndm_run_*()'` runners to target the same conda environment, set `'NDM_SOFTWARE_CONDA_ENV'` (or `'NDM_CONDA_ENV'`) once and pass that value into these helpers. The runner layer currently falls back to `'jax_cpu'` on macOS and `'ndm_software_env'` elsewhere when neither environment variable is set.

## Value

`'ndm_check_backend()'` returns `'TRUE'` invisibly when the requested environment and modules are available. It returns `'NULL'` after printing a diagnostic message when the backend is not ready.

`'ndm_build_backend()'` invisibly returns the conda environment name after attempting to provision the requested packages.

`'ndm_initialize_backend()'` returns an object of class `'ndm_backend'`. The returned list contains imported Python modules, the configured float type, and a small compatibility shim used by the legacy runtime.

`'ndm_backend_modules()'` returns the currently initialized `'ndm_backend'` object.

## Examples

```
ndm_check_backend()

## Not run:
ndm_build_backend()

## End(Not run)

## Not run:
backend <- ndm_initialize_backend()
backend$default_backend

## End(Not run)

## Not run:
ndm_initialize_backend()
ndm_backend_modules()

## End(Not run)
```

---

ndm\_create\_config

---

*Create runtime configuration objects*


---

## Description

Configuration objects collect the runtime options that are threaded through the higher-level orchestration helpers such as `'ndm_prepare_runtime()'` and `'ndm_fit()'`.

**Usage**

```
ndm_create_config(
  model_type = c("DecoderOnly", "NeuralODE"),
  backbone = "transformer",
  float_type = c("32", "64"),
  force_to_gpu = TRUE,
  resave_tfrecords = FALSE,
  gpu_mem_frac = NULL,
  ...
)

## S3 method for class 'ndm_config'
print(x, ...)
```

**Arguments**

|                               |                                                                                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>model_type</code>       | Model family metadata. Use <code>"DecoderOnly"</code> or <code>"NeuralODE"</code> .                                                                                                     |
| <code>backbone</code>         | Backbone family. Phase 1 supports <code>"transformer"</code> only.                                                                                                                      |
| <code>float_type</code>       | Floating point precision used when initializing the backend. Use <code>"32"</code> or <code>"64"</code> .                                                                               |
| <code>force_to_gpu</code>     | Logical scalar indicating whether the runtime should try to place arrays on GPU when available.                                                                                         |
| <code>resave_tfrecords</code> | Logical scalar preserved for compatibility with existing run workflows. Multidisease workflows reject <code>TRUE</code> because the legacy TFRecord-regeneration path has been retired. |
| <code>gpu_mem_frac</code>     | Optional GPU memory fraction forwarded into the runtime bootstrap code.                                                                                                                 |
| <code>...</code>              | Unused; included for S3 method compatibility.                                                                                                                                           |
| <code>x</code>                | An <code>'ndm_config'</code> object.                                                                                                                                                    |

**Value**

`'ndm_create_config()'` returns an object of class `'ndm_config'`.

`'print.ndm_config()'` prints a brief summary of an `'ndm_config'` object and invisibly returns `'x'`.

**Examples**

```
cfg <- ndm_create_config()
print(cfg)
```

---

ndm\_create\_real\_run\_config

*Package-native run configurations and workflow runners*


---

**Description**

Package-native configuration constructors and workflow runners for the self-contained real, simulation, and multidisease execution paths.



**Usage**

```

ndm_create_real_run_config(
  project_root = getwd(),
  analysis_name = "RealApril15",
  grid = NULL,
  grid_file = file.path("Data", "RunGrids", "RealGrids", sprintf("RealGrid_%s.csv",
    analysis_name)),
  outer = 3L,
  model_type = NULL,
  respect_grid_model_type = FALSE,
  resave_tfrerecords = FALSE,
  tfrecord_dir = file.path("Data", "RunTFRecords", "RealTFRecords",
    analysis_name),
  raw_data_dir = file.path("Data", "MainData"),
  outcome_metric = "inc_death",
  data_subset = "high_income",
  dry_run = FALSE
)

ndm_create_sim_run_config(
  project_root = getwd(),
  analysis_name = "BigSimsLatest",
  grid = NULL,
  grid_file = file.path("Data", "RunGrids", "SimGrids", sprintf("SimGrid_%s.csv",
    analysis_name)),
  outer = 1L,
  model_type = NULL,
  respect_grid_model_type = FALSE,
  resave_tfrerecords = TRUE,
  tfrecord_dir = file.path("Data", "RunTFRecords", "SimTFRecords",
    analysis_name),
  dry_run = FALSE
)

ndm_create_multidisease_run_config(
  project_root = getwd(),
  analysis_name = "RealLatest",
  grid = NULL,
  grid_file = file.path("Data", "RunGrids", "RealGrids", sprintf("RealGrid_%s.csv",
    analysis_name)),
  outer = 1L,
  model_type = NULL,
  respect_grid_model_type = FALSE,
  resave_tfrerecords = FALSE,
  tfrecord_dir = file.path("Data", "RunTFRecords", "RealTFRecords",
    analysis_name),
  outcome_metric = "CountValue",
  data_subset = "all",
  disease_names = c("Covid", "Flu"),
  data_format = "IHME",
  dry_run = FALSE
)

```

```
ndm_run_real(config = ndm_create_real_run_config())
```

```
ndm_run_sim(config = ndm_create_sim_run_config())
```

```
ndm_run_multidisease(config = ndm_create_multidisease_run_config())
```

## Arguments

|                                      |                                                                                                                                                                                                       |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>project_root</code>            | Working project directory used for grids, TFRecords, and outputs.                                                                                                                                     |
| <code>analysis_name</code>           | Analysis label threaded through output paths.                                                                                                                                                         |
| <code>grid</code>                    | Optional in-memory grid data frame. When supplied, <code>grid_file</code> must be <code>NULL</code> .                                                                                                 |
| <code>grid_file</code>               | Optional CSV path for the run grid.                                                                                                                                                                   |
| <code>outer</code>                   | Integer vector of outer-iteration rows to execute.                                                                                                                                                    |
| <code>model_type</code>              | Optional model family override.                                                                                                                                                                       |
| <code>respect_grid_model_type</code> | Logical scalar indicating whether grid rows may override the requested model type.                                                                                                                    |
| <code>resave_tfreCORDs</code>        | Logical scalar controlling TFRecord regeneration. Supported for real and simulation workflows. Multidisease workflows reject <code>TRUE</code> because the legacy regeneration path has been retired. |
| <code>tfrecord_dir</code>            | Optional TFRecord output directory.                                                                                                                                                                   |
| <code>raw_data_dir</code>            | Real-data input directory.                                                                                                                                                                            |
| <code>outcome_metric</code>          | Outcome metric name for real-data or multidisease runs.                                                                                                                                               |
| <code>data_subset</code>             | Optional real-data or multidisease subset name.                                                                                                                                                       |
| <code>disease_names</code>           | Character vector of multidisease names.                                                                                                                                                               |
| <code>data_format</code>             | Multidisease data format.                                                                                                                                                                             |
| <code>dry_run</code>                 | Logical scalar indicating whether the run should stop after resolving paths and grid rows.                                                                                                            |
| <code>config</code>                  | A package-native run configuration created by the matching <code>ndm_create_*_run_config()</code> helper.                                                                                             |

## Details

`dry_run = TRUE` resolves paths and selected rows without executing the underlying workflow. Dry runs can use lightweight in-memory grids such as the `BaseID` / `ModelType` preview shown in `README.md`.

Non-dry execution expects Analysis2-compatible grids. The tested simulation grids in this package are produced by `ndmdatasets::ndm_sim_build_grid()` and then augmented with fields such as `paddingMethod`, `lookahead`, `n_time_steps`, `n_inference_batches`, `scaling_batches`, and either `model_spec_name` or `model_tex_loc`. The tested real and multidisease grids include `BaseID`, `ContextLength`, `evaluationTime`, `initialTransform`, `initialNormType`, `paddingMethod`, `OSSType`, `dataInputs`, `ModelType`, `ModelDepth`, `ModelDims`, `nSamplesTrain`, `nObsInference`, `floatType`, and either `model_spec_name` or `model_tex_loc`.

Non-dry runner workflows also require the full execution stack: the `ndmdatasets` package, the helper packages used by the runtime, and a backend environment aligned with `NDM_SOFTWARE_CONDA_ENV`, `NDM_CONDA_ENV`, or the platform-specific fallback used by the runner layer.

**Value**

The `ndm_create*_run_config()` helpers return classed lists describing the requested workflow. `ndm_run_real()` and `ndm_run_sim()` return a dry-run preview when `config$dry_run` is `TRUE`, a list containing `written_base_ids`, `tfrecord_dir`, and `write_plan` when `config$resave_tfrecords` is `TRUE`, and otherwise the underlying workflow result. `ndm_run_multidisease()` returns a dry-run preview when `config$dry_run` is `TRUE` and otherwise the underlying workflow result.

**Examples**

```
## Not run:
preview_grid <- data.frame(
  BaseID = c(1L, 2L),
  ModelType = c("DecoderOnly", "NeuralODE"),
  stringsAsFactors = FALSE
)

cfg <- ndm_create_sim_run_config(
  project_root = tempdir(),
  grid = preview_grid,
  outer = 1:2,
  dry_run = TRUE
)

ndm_run_sim(cfg)

## End(Not run)
```

---

ndm\_create\_tfrecord\_parser

*Create and read TensorFlow TFRecord datasets*


---

**Description**

These wrappers construct parsers for the ndm TFRecord schema and use them to build TensorFlow datasets or eagerly collect batches into R.

**Usage**

```
ndm_create_tfrecord_parser(field_names, dtype_map = NULL, tensorflow = NULL)

ndm_read_tfrecord_dataset(
  file,
  batch_size,
  field_names,
  dtype_map = NULL,
  max_examples = NULL,
  shuffle = FALSE,
  buffer_scaler = 10L,
  tensorflow = NULL
)

ndm_collect_tfrecord_batches(
```

```

    file,
    batch_size,
    field_names,
    dtype_map = NULL,
    max_batches = NULL,
    max_examples = NULL,
    shuffle = FALSE,
    buffer_scaler = 10L,
    tensorflow = NULL
  )

```

### Arguments

|               |                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| field_names   | Character vector of TFRecord field names that should be parsed from each example.                                                                                                              |
| dtype_map     | Optional named vector or list mapping 'field_names' to TensorFlow dtype names or TensorFlow dtype objects.                                                                                     |
| tensorflow    | Optional TensorFlow module. When omitted, ndm uses the active backend's TensorFlow module when available and otherwise imports TensorFlow from the active reticulate environment at call time. |
| file          | Path to the '.tfrecord' file to read.                                                                                                                                                          |
| batch_size    | Batch size used when constructing the TensorFlow dataset.                                                                                                                                      |
| max_examples  | Optional maximum number of examples to consume from the file before batching.                                                                                                                  |
| shuffle       | Logical scalar indicating whether the dataset should be shuffled.                                                                                                                              |
| buffer_scaler | Integer multiplier used to derive the TensorFlow shuffle buffer size from 'batch_size'.                                                                                                        |
| max_batches   | Optional maximum number of batches to collect when using 'ndm_collect_tfrecord_batches()'.                                                                                                     |

### Details

TFRecord parsing requires TensorFlow to be importable from the active reticulate environment. In practice this usually means provisioning the backend with 'ndm\_build\_backend(include\_tensorflow = TRUE)' and then calling 'ndm\_initialize\_backend(import\_tensorflow = TRUE)' for the same conda environment before reading datasets.

### Value

'ndm\_create\_tfrecord\_parser()' returns a parser function compatible with 'tf\$data\$Dataset\$map()'. 'ndm\_read\_tfrecord\_dataset()' returns a TensorFlow dataset object. 'ndm\_collect\_tfrecord\_batches()' returns a list of parsed batches.

### Examples

```

fields <- ndm_tfrecord_fields_real()
parser <- ndm_create_tfrecord_parser(fields, ndm_tfrecord_dtype_map("real"))
is.function(parser)

```

---

ndm\_model\_spec\_presets

*Inspect and create epidemic model specifications*


---

## Description

The package ships a small registry of built-in compartmental model specifications and also supports importing custom ‘.tex’ model definitions from disk. The structured ‘ndm\_model\_spec’ object is the canonical representation used by the runtime wrappers.

## Usage

```
ndm_model_spec_presets()

ndm_model_spec(
  preset = NULL,
  compartments = c("seir", "seirs"),
  dynamic_beta = NULL,
  dynamic_global = FALSE,
  multi_outcome = FALSE,
  model_type = c("DecoderOnly", "NeuralODE"),
  time_varying_terms = NULL,
  endogenous_terms = NULL,
  tex_path = NULL
)

ndm_model_spec_from_tex(path, model_type = c("DecoderOnly", "NeuralODE"))

ndm_model_spec_to_tex(spec, path = NULL)

## S3 method for class 'ndm_model_spec'
print(x, ...)
```

## Arguments

|                    |                                                                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| preset             | Optional preset name from ‘ndm_model_spec_presets()’. When omitted, the remaining arguments are used to infer a built-in preset. |
| compartments       | Compartment family used when inferring a built-in preset.                                                                        |
| dynamic_beta       | Logical scalar controlling whether the local transmission rate is time-varying for built-in presets.                             |
| dynamic_global     | Logical scalar controlling whether the global rates are time-varying for built-in presets.                                       |
| multi_outcome      | Logical scalar indicating whether the observation model should include multiple outcomes.                                        |
| model_type         | Model family metadata attached to the returned specification. Either “DecoderOnly” or “NeuralODE”.                               |
| time_varying_terms | Optional character vector overriding the default time-varying term metadata stored on the returned spec.                         |

|                  |                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| endogenous_terms | Optional character vector overriding the default endogenous term metadata stored on the returned spec.             |
| tex_path         | Optional path to a custom '.tex' file. When supplied, 'ndm_model_spec()' delegates to 'ndm_model_spec_from_tex()'. |
| path             | Path to a '.tex' model specification file on disk.                                                                 |
| spec             | An 'ndm_model_spec' object.                                                                                        |
| x                | An 'ndm_model_spec' object.                                                                                        |
| ...              | Unused; included for S3 method compatibility.                                                                      |

### Value

'ndm\_model\_spec\_presets()' returns a data frame describing the built-in presets bundled with the package.

'ndm\_model\_spec()' returns an object of class 'ndm\_model\_spec'. Built-in presets include meta-data such as the packaged source path and the normalized TeX contents.

'ndm\_model\_spec\_from\_tex()' returns an 'ndm\_model\_spec' object whose 'source\_path' points at the supplied file. When the basename matches a built-in preset, the preset metadata is reused.

'ndm\_model\_spec\_to\_tex()' returns the TeX text as a single character string when 'path' is 'NULL'. When 'path' is supplied, it writes the specification to disk and invisibly returns the normalized output path.

'print.ndm\_model\_spec()' prints a brief summary of an 'ndm\_model\_spec' object and invisibly returns 'x'.

### Examples

```
ndm_model_spec_presets()

spec <- ndm_model_spec()
print(spec)

spec <- ndm_model_spec()
tex_path <- tempfile(fileext = ".tex")
ndm_model_spec_to_tex(spec, tex_path)
ndm_model_spec_from_tex(tex_path)

spec <- ndm_model_spec()
tex <- ndm_model_spec_to_tex(spec)
nchar(tex) > 0
```

---

ndm\_new\_runtime\_env     *Create and populate runtime environments*

---

### Description

These helpers manage isolated environments used to load package-managed runtime code and seed it with R values.

**Usage**

```
ndm_new_runtime_env(parent = globalenv())

ndm_set_runtime_globals(env, values, overwrite = TRUE)
```

**Arguments**

|           |                                                                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| parent    | Parent environment for a new runtime environment. Defaults to 'globalenv()' so package-managed runtime code can resolve attached-package functions consistently. |
| env       | Runtime environment that should receive new bindings.                                                                                                            |
| values    | Named list of values to assign into 'env'.                                                                                                                       |
| overwrite | Logical scalar indicating whether existing bindings in 'env' should be overwritten.                                                                              |

**Value**

'ndm\_new\_runtime\_env()' returns an environment of class 'ndm\_runtime\_env'. 'ndm\_set\_runtime\_globals()' invisibly returns 'env'.

**Examples**

```
env <- ndm_new_runtime_env()
ndm_set_runtime_globals(env, list(example_value = 1L))
env$example_value
```

---

ndm\_predict

---

*Generate predictions or evaluate the training loss*


---

**Description**

These helpers call the runtime prediction and loss functions stored in a prepared runtime environment.

**Usage**

```
ndm_predict(x, batch = NULL, inference = TRUE, seed = 1L, update_state = TRUE)

ndm_loss(
  x,
  batch = NULL,
  y = NULL,
  y_mask = NULL,
  iteration = 1L,
  seed = 1L,
  update_state = TRUE
)
```

**Arguments**

|              |                                                                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| x            | Either an 'ndm_model', an 'ndm_trained_model', or a prepared runtime environment that already contains the required runtime objects.                                                                       |
| batch        | Optional batch object. Supply either a named TFRecord-style list or the four-element packaged list returned by 'ndm_batch_to_model_inputs()'. When 'NULL', the runtime must already contain 'batch_l_cal'. |
| inference    | Logical scalar indicating whether 'ndm_predict()' should use the inference prediction path or the training prediction path.                                                                                |
| seed         | Integer seed used to derive the per-example JAX RNG keys.                                                                                                                                                  |
| update_state | Logical scalar indicating whether the runtime 'state' object should be updated with the returned state.                                                                                                    |
| y            | Optional observed outcome tensor passed to 'ndm_loss()'. When omitted, 'YTrue_out' is inferred from 'batch'.                                                                                               |
| y_mask       | Optional outcome mask tensor passed to 'ndm_loss()'. When omitted, 'YTrue_out_mask' is inferred from 'batch'.                                                                                              |
| iteration    | Training iteration number forwarded to the runtime loss function.                                                                                                                                          |

**Value**

'ndm\_predict()' returns the prediction object produced by the runtime. 'ndm\_loss()' returns the loss object produced by the runtime.

**Examples**

```
## Not run:
preds <- ndm_predict(model, batch = batch_l_cal)
loss <- ndm_loss(model, batch = batch_l_cal)

## End(Not run)
```

---

ndm\_prepare\_runtime     *Prepare a local runtime for execution*

---

**Description**

These wrappers load package-owned runtime components into an isolated environment and then source the selected data generator.

**Usage**

```
ndm_prepare_runtime(
  config = ndm_create_config(),
  runtime_env = ndm_new_runtime_env(),
  runtime_globals = list()
)

ndm_prepare_data(
  runtime_env,
  generator = c("sim", "real", "multidisease"),
  runtime_globals = list()
)
```



**Arguments**

|                              |                                                                                                        |
|------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>config</code>          | An object of class <code>'ndm_config'</code> , usually created by <code>'ndm_create_config()'</code> . |
| <code>runtime_env</code>     | Runtime environment that should receive the sourced objects.                                           |
| <code>runtime_globals</code> | Named list of additional bindings to assign before the runtime code is sourced.                        |
| <code>generator</code>       | Which data generator to source into <code>'runtime_env'</code> .                                       |

**Details**

Before building, training, predicting, or restoring artifacts from a runtime environment, initialize the backend for the target conda environment with `'ndm_initialize_backend()'`. `'ndm_prepare_runtime()'` requires the `'fastmatch'` package. `'ndm_prepare_data(generator = "sim")'` additionally requires `'progress'` and `'zoo'`.

**Value**

`'ndm_prepare_runtime()'` invisibly returns `'runtime_env'` after loading the package-owned helper and backend code.

`'ndm_prepare_data()'` invisibly returns `'runtime_env'` after sourcing the requested data generator.

**Examples**

```
env <- ndm_new_runtime_env()
cfg <- ndm_create_config()
## Not run:
ndm_prepare_runtime(cfg, env)

## End(Not run)

env <- ndm_new_runtime_env()
ndm_set_runtime_globals(env, list(example_value = 1L))
```

---

ndm\_print

---

*Print a timestamped diagnostic message*


---

**Description**

`'ndm_print()'` is a small logging helper used throughout the package-facing orchestration code.

**Usage**

```
ndm_print(text, quiet = FALSE)
```

**Arguments**

|                    |                                                                |
|--------------------|----------------------------------------------------------------|
| <code>text</code>  | Text to print.                                                 |
| <code>quiet</code> | Logical scalar indicating whether output should be suppressed. |

**Value**

The input `'text'`, invisibly.

## Examples

```
ndm_print("starting")
```

---

|                |                                         |
|----------------|-----------------------------------------|
| ndm_save_model | <i>Save and restore model artifacts</i> |
|----------------|-----------------------------------------|

---

## Description

These helpers persist the public ‘ndm\_model’ / ‘ndm\_trained\_model’ objects to a versioned artifact directory, restore them into a fresh runtime environment, and resume training from the restored optimizer state.

## Usage

```
ndm_save_model(x, path, bundle = NULL, overwrite = FALSE)
```

```
ndm_load_model(
  path,
  bundle = NULL,
  calibration_source = c("inference", "train")
)
```

```
ndm_resume_training(
  path,
  bundle = NULL,
  calibration_source = c("train", "inference"),
  train_globals = list()
)
```

## Arguments

|                    |                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|
| x                  | An object of class ‘ndm_model’ or ‘ndm_trained_model’.                                                                      |
| path               | Artifact directory path.                                                                                                    |
| bundle             | Optional ‘ndm_tfrecord_bundle’ used to record TFRecord references for later resume workflows.                               |
| overwrite          | Logical scalar indicating whether an existing artifact directory should be replaced.                                        |
| calibration_source | Which dataset should supply the preview batch used to seed runtime globals such as ‘batch_1_cal’ when ‘bundle’ is attached. |
| train_globals      | Optional named list of runtime globals applied immediately before ‘ndm_resume_training()’ continues the training loop.      |

## Details

These helpers assume ‘ndm\_initialize\_backend()’ has already run for the active conda environment. ‘ndm\_save\_model()’ only works for runtime-backed ‘ndm\_model’ / ‘ndm\_trained\_model’ objects whose environments still contain the built model state, including ‘ModelList’ and ‘state’. Exact resume also requires saved optimizer state. When ‘bundle = NULL’ in ‘ndm\_resume\_training()’, the artifact metadata must already contain a recorded TFRecord bundle reference. ‘ndm\_load\_model()’ and training-restore paths require the ‘rrapply’ package.

**Value**

‘ndm\_save\_model()’ returns the normalized artifact directory path, invisibly. ‘ndm\_load\_model()’ returns an ‘ndm\_model’ or ‘ndm\_trained\_model’ matching the saved artifact. ‘ndm\_resume\_training()’ returns an ‘ndm\_trained\_model’.

**Examples**

```
## Not run:
# After ndm_initialize_backend() and ndm_train() / ndm_fit():
artifact_dir <- ndm_save_model(trained_model, tempfile("ndm-artifact-"))
restored <- ndm_load_model(artifact_dir, bundle = tf_bundle)
resumed <- ndm_resume_training(artifact_dir, bundle = tf_bundle)

## End(Not run)
```

---

ndm\_tfrecord\_fields\_real

*Inspect TFRecord field contracts*


---

**Description**

These helpers describe the field names and default TensorFlow dtypes expected by the packaged TFRecord readers.

**Usage**

```
ndm_tfrecord_fields_real()

ndm_tfrecord_fields_sim()

ndm_tfrecord_dtype_map(kind = c("real", "sim"))
```

**Arguments**

|      |                                                                                                    |
|------|----------------------------------------------------------------------------------------------------|
| kind | TFRecord schema to inspect. Use "real" for observed-data records and "sim" for simulation records. |
|------|----------------------------------------------------------------------------------------------------|

**Value**

‘ndm\_tfrecord\_fields\_real()’ and ‘ndm\_tfrecord\_fields\_sim()’ return character vectors of field names. ‘ndm\_tfrecord\_dtype\_map()’ returns a named character vector mapping each field to its default TensorFlow dtype.

**Examples**

```
ndm_tfrecord_fields_real()
ndm_tfrecord_dtype_map("sim")
```

---

ndm\_tf\_batch\_to\_r      *Convert and bundle TFRecord batches*


---

## Description

These helpers convert TensorFlow batches to native R or JAX objects, package them into the shape expected by the package-managed runtime, and attach dataset handles to a runtime environment.

## Usage

```
ndm_tf_batch_to_r(batch)

ndm_tf_batch_to_jax(batch, backend = NULL)

ndm_batch_to_model_inputs(batch)

ndm_load_tfrecord_bundle(
  train_file,
  inference_file = NULL,
  batch_size,
  kind = c("real", "sim"),
  dtype_map = NULL,
  tensorflow = NULL,
  max_train_examples = NULL,
  max_inference_examples = NULL,
  shuffle_train = TRUE,
  shuffle_inference = FALSE
)

ndm_attach_tfrecord_bundle(
  env,
  bundle,
  backend = NULL,
  calibration_source = c("inference", "train")
)
```

## Arguments

|                |                                                                                                                                                                                                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| batch          | A parsed TFRecord batch or a nested list containing TensorFlow tensors.                                                                                                                        |
| backend        | Optional ‘ndm_backend’ object. When omitted, ‘ndm_tf_batch_to_jax()’ and ‘ndm_attach_tfrecord_bundle()’ use the active backend from ‘ndm_backend_modules()’ when available.                    |
| train_file     | Path to the training TFRecord file.                                                                                                                                                            |
| inference_file | Optional path to an inference TFRecord file.                                                                                                                                                   |
| batch_size     | Batch size used when constructing the training and inference datasets.                                                                                                                         |
| kind           | TFRecord schema to use when reading ‘train_file’ and ‘inference_file’.                                                                                                                         |
| dtype_map      | Optional named dtype mapping passed through to the TFRecord readers.                                                                                                                           |
| tensorflow     | Optional TensorFlow module. When omitted, ndm uses the active backend’s TensorFlow module when available and otherwise imports TensorFlow from the active reticulate environment at call time. |

|                                     |                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>max_train_examples</code>     | Optional cap on the number of training examples to read.                                                        |
| <code>max_inference_examples</code> | Optional cap on the number of inference examples to read.                                                       |
| <code>shuffle_train</code>          | Logical scalar indicating whether the training dataset should be shuffled.                                      |
| <code>shuffle_inference</code>      | Logical scalar indicating whether the inference dataset should be shuffled.                                     |
| <code>env</code>                    | Runtime environment that should receive dataset handles and the preview calibration batch.                      |
| <code>bundle</code>                 | An object of class <code>'ndm_tfrecord_bundle'</code> returned by <code>'ndm_load_tfrecord_bundle()'</code> .   |
| <code>calibration_source</code>     | Which dataset should supply the preview batch used to seed runtime globals such as <code>'batch_1_cal'</code> . |

### Value

`'ndm_tf_batch_to_r()'` recursively converts a TensorFlow batch to R arrays and lists. `'ndm_tf_batch_to_jax()'` recursively converts tensors into JAX arrays. `'ndm_batch_to_model_inputs()'` returns the packaged four-element list expected by the package-managed model stages. `'ndm_load_tfrecord_bundle()'` returns an `'ndm_tfrecord_bundle'` object. `'ndm_attach_tfrecord_bundle()'` invisibly returns `'env'`.

### Examples

```
batch <- list(
  XPred = array(0, c(2, 3, 4)),
  XPred_mask = array(TRUE, c(2, 3, 4)),
  Context = array(0, c(2, 1, 4)),
  Context_mask = array(TRUE, c(2, 1, 4)),
  location_id_numeric = matrix(1L, nrow = 2),
  time_id_numeric = matrix(1L, nrow = 2)
)
packaged <- ndm_batch_to_model_inputs(batch)
length(packaged)
```

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